

Assignment-1

Number System

1) complete the table with equivalents in binary, octal and hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hexa)

Decimal	binary	octal	hexa
143.75			
	11011.101		
		453	
			1F.C

→ (a) $(143.75)_{10} \rightarrow (?)_2$ decimal → binary

$$\begin{array}{r}
 2 \overline{) 143} \\
 \underline{2 71} - 1 \\
 2 \overline{) 35} - 1 \\
 2 \overline{) 17} - 1 \\
 2 \overline{) 8} - 1 \\
 2 \overline{) 4} - 0 \\
 2 \overline{) 2} - 0 \\
 1 - 0
 \end{array}$$

$0.75 \times 2 = 1.5$
 $0.5 \times 2 = 1$
 $0.25 \times 2 = 0.5$
 $0.5 \times 2 = 1$

$$\therefore (143.75)_{10} = (10001111.11)_2$$

$$143 = 10001111$$

$$(10001111.11)_2 \rightarrow (?)_8$$

binary → octal

$$10001111.11$$

$$\underline{010} \underline{001} \underline{111} \underline{110}$$

$$217.6$$

$$\therefore (10001111.11)_2 = (217.6)_8$$

$$(10001111.11)_2 \rightarrow (?)_{16}$$

binary → hexa

$$10001111.11$$

$$\underline{1000} \underline{1111} \underline{1100}$$

$$8F.C$$

$$\therefore (10001111.11)_2 = (8F.C)_{16}$$

$$(b) (11011.101)_2 \rightarrow (?)_{10}$$

binary → decimal

$$= 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 + 1 \times \frac{1}{2} + 0 \times \frac{1}{4} + 1 \times \frac{1}{8}$$

$$= 16 + 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$$

$$= 27.625$$

$$\therefore (11011.101)_2 = (27.625)_{10}$$

$$(11011.101)_2 \rightarrow (?)_8$$

binary $\rightarrow$ octal

$$11011.101$$

$$\begin{array}{ccc} 011 & 011 & 101 \\ \hline 3 & 3 & 5 \end{array}$$

$$\therefore (11011.101)_2 = (33.5)_8$$

$$(11011.101)_2 \rightarrow (?)_{16}$$

binary $\rightarrow$ hex

$$11011.101$$

$$\begin{array}{ccc} 0001 & 1011 & 1010 \\ \hline 1 & B & A \end{array}$$

$$\therefore (11011.101)_2 = (1B.A)_{16}$$

$$(c) (453)_8 \rightarrow (?)_{10}$$

octal $\rightarrow$ decimal

$$= 4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0$$

$$= 256 + 40 + 3$$

$$= 299$$

$$\therefore (453)_8 = (299)_{10}$$

$$(453)_8 \rightarrow (?)_2$$

octal $\rightarrow$ binary

$$4 \rightarrow 100$$

$$5 \rightarrow 101$$

$$3 \rightarrow 011$$

$$100 \ 101 \ 011$$

$$\therefore (453)_8 = (100101011)_2$$

$$(100101011)_2 \rightarrow (?)_{16}$$

binary $\rightarrow$ hex

$$100101011$$

$$\begin{array}{ccc} 0001 & 0010 & 1011 \\ \hline 1 & 2 & B \end{array}$$

$$\therefore (100101011)_2 = (12B)_{16}$$

$$(d) (1F.C)_{16} \rightarrow (?)_{10}$$

hex $\rightarrow$ decimal

$$= 1 \times 16^1 + 15 \times 16^0 + 12 \times 16^{-1}$$

$$= 31.75$$

$$\therefore (1F.C)_{16} = (31.75)_{10}$$

$$(1F.C)_{16} \rightarrow (?)_2$$

hex $\rightarrow$ binary

$$1 \rightarrow 0001$$

$$F \rightarrow 1111$$

$$C \rightarrow 1100$$

$$0001 \ 1111 \ 1100$$

$$1111.11$$

$$\therefore (1F.C)_{16} = (1111.11)_2$$

$$(11111.11)_2 \rightarrow (?)_8$$

binary $\rightarrow$ octal

$$11111.11$$

$$\begin{array}{r} 011 \quad 1111 \quad 110 \\ \hline 37.6 \end{array}$$

$$\therefore (11111.11)_2 = (37.6)_8$$

Final Table

Decimal	Binary	Octal	hex
143.75	10001111.11	217.6	8F.C
27.625	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	11111.11	37.6	1F.C

2] Find Equivalents

(a) $(324)_5 = (?)_{10}$

$$\rightarrow = (3 \times 5^2) + (2 \times 5^1) + (4 \times 5^0)$$

$$= 75 + 10 + 4$$

$$= 89$$

$$\therefore (324)_5 = (89)_{10}$$

(b) $(32.23)_4 = (?)_{10}$

$$\rightarrow = (3 \times 4^1) + (2 \times 4^0) + (2 \times 4^{-1}) + (3 \times 4^{-2})$$

$$= 12 + 2 + 0.5 + 0.1875$$

$$= 14.6875$$

$$\therefore (32.23)_4 = (14.6875)_{10}$$

(c) $(231)_{10} = (?)_7$

$$\begin{array}{r} 7 \overline{)231} \\ 7 \overline{)33} - 0 \\ \quad 4 - 5 \end{array}$$

$$450$$

$$\therefore (231)_{10} = (450)_7$$

(d) $(189)_{10} = (?)_6$

$$\begin{array}{r} 6 \overline{)189} \\ 6 \overline{)31} - 3 \\ \quad 5 - 1 \end{array}$$

$$\therefore (189)_{10} = (513)_6$$

3) Find the radix r

$$(a) \left(\frac{312}{20} \right)_r = (13.1)_r$$

$$\rightarrow (312)_r = 3r^2 + 1r + 2 = 3r^2 + r + 2$$

$$(20)_r = 2r + 0 = 2r$$

$$(13.1)_r = 1r + 3 + 1r^{-1} = r + 3 + \frac{1}{r}$$

$$\therefore \frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$3r^2 - 2r^2 + r - 6r + 2 - 2 = 0$$

$$r^2 - 5r = 0$$

$$r(r - 5) = 0$$

since, radix can't be zero, $r = 5$

$$(b) 264r + 136r = 433r$$

$$\rightarrow (264)_r = 2r^2 + 6r + 4$$

$$(136)_r = 1r^2 + 3r + 6$$

$$(433)_r = 4r^2 + 3r + 3$$

$$(2r^2 + 6r + 4) + (r^2 + 3r + 6) = (4r^2 + 3r + 3)$$

$$3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$-3r^2 + 4r^2 - 9r + 3r - 10 + 3 = 0$$

$$r^2 - 6r - 7 = 0$$

$$(r - 7)(r + 1) = 0$$

since, radix can't be negative, $r = 7$

$$(c) (221505)_8 = (12345)_r$$

$$\rightarrow (221505)_8 = (2 \times 8^5) + (2 \times 8^4) + (1 \times 8^3) + (5 \times 8^2) + (0 \times 8^1) + (5 \times 8^0)$$

$$= 65536 + 8192 + 512 + 320 + 0 + 5$$

$$= 74565$$

$$(12345)_r = 1r^4 + 2r^3 + 3r^2 + 4r^1 + 5r^0$$

$$= r^4 + 2r^3 + 3r^2 + 4r + 5$$

for $r=16$

$$\begin{aligned} &= 16^4 + (2 \times 16^3) + (3 \times 16^2) + (4 \times 16) + 5 \\ &= 65536 + 8192 + 768 + 64 + 5 \\ &= 74565 \end{aligned}$$

hence, $\boxed{r=16}$

4) Addition

(a) $(1101)_2 + (11011111)_2$

→

$$\begin{array}{r} 1111 \\ 00001101 \\ 11011111 \\ \hline 11101100 \end{array}$$

$$\therefore (1101)_2 + (11011111)_2 = (11101100)_2$$

(b) $(77652)_8 + (76543)_8$

→

$$\begin{array}{r} 77652 \\ 76543 \\ \hline \end{array} +$$

$\begin{array}{r} 8 \overline{)9} \\ 1-1 \end{array}$

$\begin{array}{r} 8 \overline{)12} \\ 1-4 \end{array}$

$\begin{array}{r} 8 \overline{)14} \\ 1-6 \end{array}$

$\begin{array}{r} 8 \overline{)15} \\ 1-7 \end{array}$

• $2 + 3 = 5$

• $5 + 4 = (9)_{10} = (11)_8 \rightarrow \text{write} = 1, \text{carry} = 1$

• $6 + 5 + 1 = (12)_{10} = (14)_8 \rightarrow \text{write} = 4, \text{carry} = 1$

• $7 + 6 + 1 = (14)_{10} = (16)_8 \rightarrow \text{write} = 6, \text{carry} = 1$

• $7 + 7 + 1 = (15)_{10} = (17)_8 \rightarrow \text{write} = 7, \text{carry} = 1$

• bring down the final carry 1

∴

$$\begin{array}{r} 77652 \\ 76543 \\ \hline 176415 \end{array}$$

$$\therefore (77652)_8 + (76543)_8 = (176415)_8$$

(c) $(578A)_{16} + (ABDE)_{16}$

→

$$\begin{array}{r} 578A \\ ABDE \\ \hline \end{array} +$$

$\begin{array}{r} 16 \overline{)24} \\ 1-8 \end{array}$

$\begin{array}{r} 16 \overline{)22} \\ 1-6 \end{array}$

$\begin{array}{r} 16 \overline{)19} \\ 1-3 \end{array}$

$\begin{array}{r} 16 \overline{)16} \\ 1-0 \end{array}$

• $A + E = 10 + 14 = (24)_{10} = (18)_{16} \rightarrow \text{write} = 8, \text{carry} = 1$

• $8 + D + 1 = 8 + 13 + 1 = (22)_{10} = (16)_{16} \rightarrow \text{write} = 6, \text{carry} = 1$

• $7 + B + 1 = 7 + 11 + 1 = (19)_{10} = (13)_{16} \rightarrow \text{write} = 3, \text{carry} = 1$

- $5 + A + 1 = 5 + 10 + 1 = (16)_{10} = (10)_{16} \rightarrow \text{write} = 0, \text{carry} = 1$
- Bring down the final carry 1.

$$\begin{array}{r} 578A \\ + ABDE \\ \hline 10368 \end{array}$$

$$\therefore (578A)_{16} + (ABDE)_{16} = (10368)_{16}$$

(d) $(234)_6 + (315)_6$

$$\rightarrow \begin{array}{r} 234 \\ + 315 \\ \hline \end{array}$$

- $4 + 5 = (9)_{10} = (13)_6 \rightarrow \text{write} = 3, \text{carry} = 1$
- $3 + 1 + 1 = (5)_{10} = (5)_6 \rightarrow \text{write} = 5, \text{carry} = 0$
- $2 + 3 = (5)_{10} = (5)_6 \rightarrow \text{write} = 5, \text{carry} = 0$

$$\begin{array}{r} 234 \\ + 315 \\ \hline 553 \end{array}$$

$$\therefore (234)_6 + (315)_6 = (553)_6$$

5) Binary $\rightarrow$ Gray code

(a) $(10101)_2$

• 1st gray bit = 1st binary bit

$\rightarrow$ First bit = 1

• Every next gray bit = xor of the

• $1 \oplus 0 = 1$

current binary bit & previous binary bit

• $0 \oplus 1 = 1$

Binary 1 0 1 0 1

• $1 \oplus 0 = 1$

Gray 1 1 1 1 1

• $0 \oplus 1 = 1$

(b) $(01110)_2$

$\rightarrow$ First bit = 0

• $0 \oplus 1 = 1$

• $1 \oplus 1 = 0$

Binary 0 1 1 1 0

• $1 \oplus 1 = 0$

Gray 0 1 0 0 1

• $1 \oplus 0 = 1$

6) Gray code $\rightarrow$ Binary

(a) 01101 (gray)

$\rightarrow$ First bit = 0

$$0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

• 1st binary bit = 1st gray bit

• next binary bit = previous binary

bit XOR current gray bit

Gray 0 1 1 0 1

Binary 0 1 0 0 1

(b) 10101 (gray)

$\rightarrow$ First bit = 1

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

Gray 1 0 1 0 1

Binary 1 1 0 0 1